

Vijayaragavan S

CLOUD / SOFTWARE ENGINEER

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Computer Science undergraduate graduating in 2027, with hands-on AWS application deployment and containerization experience. Built a rental platform spanning EC2, S3 and CloudFront, with API health checks and database startup checks. Also develops compact ML applications for offline deployment.

TECHNICAL SKILLS

Cloud and infrastructure: AWS EC2, S3, CloudFront, SES, IAM; Docker, Docker Compose, Linux

Application operations: Nginx reverse proxy, PM2, health endpoints, request logging, Git

Backend and databases: Node.js, Express.js, Flask, REST APIs, PostgreSQL, Prisma, SQL

Languages and deployment tools: Python, JavaScript, TypeScript, Java, LiteRT, PyTorch

PROJECTS

CampusRent | AWS Application Deployment

2025-Present | AWS, Node.js, PostgreSQL, Docker, Nginx, PM2 | Live application

- Deployed a React frontend through S3/CloudFront and an Express backend on EC2, using Nginx to proxy API requests and WebSocket upgrades for Socket.IO.
- Integrated S3 image storage and SES verification emails; used PM2 for backend process management and automatic restarts.
- Configured Compose services with persistent PostgreSQL storage, a pg_isready health check and database-ready startup ordering; applied Prisma migrations at backend startup.
- Added an API health endpoint exposing uptime and memory usage, HTTP request logging, origin restrictions and configurable rate limits.

AnaemiaScreen | Model Packaging and Offline Deployment

2026 | Python, PyTorch, LiteRT, Java, Android

- Packaged a U-Net segmenter and calibrated classifier ensembles in an offline Android app, removing the need for a network call during screening.
- Reduced a classifier from **8.57 MB to 2.54 MB** with mixed-precision quantization while retaining 0.866 AUC in a 925-subject out-of-fold comparison.
- Reduced the deployed three-model ensemble from **25.70 MB to 7.61 MB**; checked Python/Java pipeline parity across **634 scored** cases with no decision differences.

Carbon Emissions Intelligence Platform

2025-Present | Python, Flask, PostgreSQL, Docker | Source code

- Containerized an emissions analytics platform with a Flask API and PostgreSQL data layer for repeatable setup and deployment.
- Built data ingestion and validation workflows, with API-accessible analytics and Plotly dashboards for emissions reporting.

PUBLICATION

Vijayaragavan Sivasamy et al. (first author). Real-Time Pothole Detection Using Dual Accelerometer Sensors with Temporal Convolutional Networks and Attention Mechanisms. IEEE conference publication, 2026.

DOI: 10.1109/RAEEUCCI67649.2026.11504540

EDUCATION

B.Tech, Computer Science and Engineering | 2023-2027 (Expected)

SRM Institute of Science and Technology, Chennai | **CGPA: 9.46/10**

Relevant coursework: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks

Certifications: Java - NPTEL | SAP Backend ABAP Cloud Certificate